

The PreDoc logit experiment, reproduced four additions to TrustRegionRadius.jl, and what the runs say

Abstract

The variable-sampling experiment of the PreDoc needed four things the package did not have: a multiple-alternative logit, a two-sided smoothing of the sample size, the outer-product estimate of the paired variance, and the three sample management schemes. All four are now in the package, and this report gives what was added, why each was needed, how each was checked, and what the reproduction shows. The experiment is the PreDoc's: five alternatives, $p = 10$, $N = 100\,000$ individuals, $|\mathcal{S}_0| = 100$, $z_{0.95}$, and the trust-region constants $(\eta_1, \eta_2, \gamma_1, \gamma_2, \gamma_3) = (0.01, 0.8, 0.5, 0.9, 1.5)$. The central claim reproduces: the common-random-variable schemes converge in 26 and 28 iterations where independent draws do not converge in 300. Three qualifications came out of the runs that the iteration counts hide. All three arms spend within 6% of the same number of term evaluations, so the right statement is that one arm converged on the budget the other spent without converging. The mechanism the PreDoc offers for the effect is not confirmed by the natural diagnostic, which ranks the schemes in the wrong order. And the overfitting hazard the PreDoc attaches to VAC is present and measurable, at 4 of 8 sample-size falls taken with the true gradient rising, against 1 of 6 under VAI/VAC.

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1 What this supersedes

An earlier report, `report/task14-predoc-logit/`, recorded the same experiment attempted under the constraint that nothing in the package be modified. It reported four departures from the PreDoc as gaps and reported one comparison as refuted rather than made. That report is a correct record of what was possible then and is superseded by this one on every point where the two disagree. The four gaps are now features; the refuted comparison is now made directly, in Section 6.3.

2 The experiment, reformulated

2.1 The problem

Individual n chooses among J alternatives. Alternative j carries an attribute vector $x_{nj} \in \mathbb{R}^p$ and a representative utility linear in the parameters, $V_{nj} = \beta^\top x_{nj}$. Under independent Gumbel errors the choice probabilities take the closed form

$$P_{nj} = \frac{e^{\beta^\top x_{nj}}}{\sum_l e^{\beta^\top x_{nl}}}, \quad (1)$$

and maximum likelihood over a population of N individuals is the minimisation of

$$f(\beta) = -\frac{1}{N} \sum_{n=1}^N \sum_{j=1}^J y_{nj} \ln P_{nj}, \quad y_{nj} = 1 \text{ iff } n \text{ chose } j. \quad (2)$$

The PreDoc takes $J = 5$, $p = 10$ and $N = 100\,000$, on data generated from the model itself so that the information identity holds and the outer-product Hessian is justified.

2.2 The method

The method is a trust-region algorithm on a Monte-Carlo model $m_k(s) = \tilde{f}_k + \tilde{g}_k^\top s + \frac{1}{2} s^\top \tilde{B}_k s$, with three additions to the standard loop.

The sample size candidate. From a confidence statement about the achieved decrease, the PreDoc's Step 5 asks for

$$N_k^+ = \left\lceil \frac{\sigma_k^2 (\Phi^{-1}(1 - \beta_k))^2}{\tilde{\mu}_k^2} \right\rceil, \quad \tilde{\mu}_k = \tilde{f}_k(x_k + s_k) - \tilde{f}_k(x_k). \quad (3)$$

The sample update function. Step 6 passes that candidate through Ψ_k before it becomes N_{k+1} . The *naive smoothing*, written $\text{Naive}(a, b)$, is the two-sided bound

$$aN_k \leq N_{k+1} \leq bN_k, \quad 0 < a \leq 1, \quad b > 1, \quad (4)$$

alongside a floor that rises by a fixed increment, $N_{k+1}^{\min} = N_k^{\min} + N_{\text{incr}}$.

The sample management scheme. How $\mathcal{N}_k$ becomes $\mathcal{N}_{k+1}$ is a separate choice from how large it is, and the PreDoc gives three answers. The names differ between its method section and its figures, and both are kept here because both appear in the source.

figures	method	what happens between iterations
VAI	IRV	redraw independently of the previous batch
VAI/VAC	I/CRV	on a rise, keep the batch and append fresh draws, $\mathcal{N}_{k+1} = \mathcal{N}_k \cup \{\xi_{N_k+1}, \dots\}$; on a fall, keep a uniform subsample, $\mathcal{N}_{k+1} \subset \mathcal{N}_k$
VAC	CRV	the first N_k individuals, $\mathcal{N}_k = \{\xi_1, \dots, \xi_{N_k}\}$; finite population only

Which variance. Equation (3) needs σ_k^2 , and the PreDoc offers two, labelling its runs *tv* and *sHs*. The first is the empirical variance of the individual decreases. The second comes from a first-order expansion of $F(\cdot, \xi)$ about x_k , under which $\text{Var}[F(x_k + s_k, \xi) - F(x_k, \xi)] \approx s_k^\top \text{Var}[\nabla F(x_k, \xi)] s_k$, and replacing $\text{Var}[\nabla F]$ by $\tilde{B}_k - \tilde{g}_k \tilde{g}_k^\top$ gives

$$\hat{\sigma}_{N_k}^2 \approx s_k^\top \tilde{B}_k s_k - (\tilde{g}_k^\top s_k)^2. \quad (5)$$

Its appeal is that both terms are already formed: $s_k^\top \tilde{B}_k s_k$ is needed for ρ_k and $\tilde{g}_k$ is in memory for the model.

3 Audit: what the package already had

Five of the eight components needed no work.

PreDoc	package	note
Step 5, equation (3)	CertifiedDecrease	on paired differences
Step 3, truncated CG	SteihaugCG	
Step 7, equation for Δ_{k+1}	RDelta	$\gamma_1, \gamma_2, \gamma_3$ as given
$\tilde{B}_k$, the OP approximation	BHHHModel	
a subsampled true Hessian	ExactHessian	the PreDoc's <i>TH</i>

CertifiedDecrease is worth a word because the correspondence is exact and not obvious. It sizes the batch from the sample mean and variance of the paired differences $D_i = F(x_k, \xi_i) - F(x_k + s_k, \xi_i)$, taken under common random numbers, which is (3) with the sign of $\tilde{\mu}_k$ flipped so the mean decrease is positive. Pairing is not an optimisation: with independent draws at the two ends of the step, $\text{Var}(D_i)$ does not shrink as $\|s_k\| \rightarrow 0$, and the rule becomes unusable near a solution.

Three thresholds also need care. The package separates acceptance from radius scaling: acceptance is decided by η and scaling by η_1 and η_2 , whereas the PreDoc accepts on its own η_1 . Setting $\eta = \eta_1 = 0.01$ reproduces it.

The confidence level is passed as p and the quantile derived from it. The notebook checks that this reaches the PreDoc's constant rather than assuming it: **CertifiedDecrease**($p = 0.95$).z is 1.6448534923 against the quoted $z_{0.95} \approx 1.6449$, a difference of 4.651×10^{-5} , which is agreement to the four digits quoted.

4 The four additions

4.1 MultinomialLogit

A **LikelihoodProblem** carrying the $M \times J \times p$ attribute array and the chosen alternative. It implements (1) and (2) directly. Writing $\bar{x}_n = \sum_j P_{nj} x_{nj}$ for the probability-weighted mean attribute, the per-observation score and Hessian are

$$s_n = \bar{x}_n - x_{nc_n}, \quad H_n = \sum_j P_{nj} x_{nj} x_{nj}^\top - \bar{x}_n \bar{x}_n^\top, \quad (6)$$

so H_n is the covariance of the attribute vector under the choice probabilities. It is positive semidefinite, and the objective is therefore convex. This matters downstream: `BHHHModel` is positive semidefinite by construction and so can never report negative curvature, and on a convex problem that costs nothing. The comparison against the exact Hessian in Section 6.4 is therefore about approximation quality alone.

The default population is the PreDoc’s $100\,000 \times 5 \times 10$, which is 40 MB.

The generating noise, and a point we do not settle. The PreDoc says that during generation “some noise is injected in the data by adding some perturbation, with mean zero, to the true model parameters β^* ”, and concludes that the information identity holds. Read literally, a separate $\beta_n = \beta^* + \varepsilon_n$ per individual is a *mixed* logit generating process fitted with a plain logit, which is misspecification, and the identity then fails at every β and does not improve with M .

The keyword `noise` implements the literal reading, and measurement settles what it costs: at β^* with $M = 4000$, the BHHH identity error is 0.064473 at `noise = 0` and 1.365955 at `noise = 1.5`. The runs in this report all use `noise = 0`, the correctly specified case, so that the premise Table 1 checks is the premise the runs rely on. We have not reconciled the two readings and do not claim to.

4.2 SmoothedSize

A wrapper over any `SamplingRule`, following the package’s own idiom for composition. The inner rule proposes N_{k+1}^+ and the wrapper reports

$$N_{k+1} = \min\{bN_k, \max\{aN_k, N_k^{\min}, N_{k+1}^+\}\}, \quad N_{k+1}^{\min} = N_k^{\min} + N_{incr}, \quad (7)$$

which is (4) together with the PreDoc’s floor. Every trait is forwarded, so the wrapper is transparent to the problem-compatibility checks, to the oracle’s cap logic and to the solver.

Two design points are worth recording. The gradient and objective channels are smoothed separately against their own previous sizes, so a rule that distinguishes them keeps the distinction; a rule that returns one size for both, which is every rule that pairs, is unaffected. And when the rising floor exceeds the ceiling bN_k , which happens most easily just after a sharp fall, the floor wins: it is a requirement the caller stated outright, where the ceiling is a smoothing preference.

This subsumes what was there before. `CertifiedDecrease(monotone = true)` is the special case $a = 1$, and `monotone = false` is $a = 0, b = \infty$. Neither is the PreDoc’s setting, and Table 2 shows the two extremes behaving quite differently from `Naive(0.75, 2)`.

4.3 CertifiedDecrease(variance = ...)

A keyword taking `:empirical` (the default, unchanged) or `:outer_product`, together with a trait `paired_variance_kind` that the solver reads to decide what to hand the rule. The outer-product estimate is computed by `paired_op_variance`.

It is formed as the sample variance of the scalars $\nabla F_i^\top s_k$ rather than as the difference $s^\top \tilde{B}s - (\tilde{g}^\top s)^2$ of two quadratic forms. The two are algebraically the same, and the first cannot come out negative when the two terms nearly cancel. This is not a hypothetical concern: it is exactly the failure the earlier report had to record when it tried to reconstruct the quantity after the fact from an archived trace.

Both estimates are now recorded on the trace whenever the problem supplies scores, under

`:paired_variance_trajectory` and `:paired_op_variance_trajectory`,

so the estimate a run did not use is auditable against the one it did, on the same batch at the same iterate. The cost is one score matrix over the objective batch per paired iteration.

One distinction is easy to lose and worth stating. The outer product appears here as an estimate of $\text{Var}[\nabla F]$, not as an approximation to the Hessian. It does *not* require the information identity, and it remains legitimate on a misspecified model where BHHH as a Hessian is not.

4.4 FiniteSumNLP(scheme = ...)

A field on both sampled oracles, taking `:independent`, `:nested` or `:prefix`, implementing IRV, I/CRV and CRV to the definitions quoted above. Within an iteration the batch is fixed and every evaluation shares it, which is untouched; the scheme decides only what happens between iterations.

`:prefix` needs a finite population and is rejected on an expectation, which is the PreDoc’s own restriction of CRV. It consumes no random numbers at all, which the notebook checks by comparing the oracle’s generator state across five resamples. `:nested` is supported on both classes through two small functions the problem type can extend, with methods for index vectors and for Gaussian draws.

5 How the four were checked

Forty-five checks, each against the definition rather than against the implementation. The full suite of 1452 pre-existing package tests also passes unchanged, which is what establishes that the defaults are untouched.

feature	check	result
MultinomialLogit	gradient against central differences of the objective	rel. 5.05×10^{-10}
	Hessian against central differences of the gradient	rel. 5.74×10^{-10}
	Hessian positive semidefinite	$\lambda_{\min} = 0.670$
	$f(0) = \ln J$ over the whole population	6.66×10^{-16}
	mean of the scores equals the gradient	0
SmoothedSize	every transition of a 300-iteration run inside (4)	0 violations
	the same run’s floor $10 + 10k$ respected	0 violations
	the bare rule does leave those bounds	ratios in $[0.10, 7.87]$
variance	paired_op_variance equals $s^\top \tilde{B}s - (\tilde{g}^\top s)^2$	rel. $< 10^{-9}$
	the two estimates agree as $\ s\ \rightarrow 0$	ratio $1.0059 \rightarrow 1.0000$
	rejected on a problem with no scores	ArgumentError
schemes	:prefix batch is $1:N_k$ at every iteration	holds
	:prefix consumes no randomness	generator unchanged
	:nested keeps the batch on a rise	$\mathcal{N}_k \subseteq \mathcal{N}_{k+1}$
	:nested subsets it on a fall	$\mathcal{N}_{k+1} \subset \mathcal{N}_k$
	:independent does <i>not</i> nest	confirmed

The last of those matters as much as the others: without it the nesting checks would pass on a scheme that nested by accident, and would be evidence of nothing.

6 Results

Every arm below carries the PreDoc’s trust-region constants, $x_0 = 0$, $\Delta_0 = 0.1 \|\tilde{g}_0\|$ with $\tilde{g}_0$ over the initial batch of $|\mathcal{S}_0| = 100$, and the same seed. Arms differ only in what they are named for. Two settings are ours and not the PreDoc’s, which states neither: a cap of 300 iterations and a gradient tolerance of 10^{-6} .

Every run is scored after the fact by the same code, on the true objective and true gradient over all 100 000 individuals at the point it stopped, never on the solver’s own stopping test. An arm reporting `max_iter` is reported as not having converged, not as having taken that many iterations.

6.1 The premise: does the information identity hold?

Table 1: The information identity, measured over all 100 000 individuals. B_{err} is the BHHH relative error, W_{err} the BHHH-2 one, and the gap between them is the rank-one term that vanishes at a stationary point.

point	B_{err}	W_{err}	BW_{gap}	$\ g\ $
β^* (the truth)	0.011950	0.011949	0.000010	3.5033×10^{-3}
$\beta^* + 0.5$	0.832126	0.599609	0.342115	6.7378×10^{-1}
0 (the PreDoc’s x_0)	0.245365	0.157656	0.402691	1.0094×10^0

The identity holds at β^* to 1.2%, and the residual is the $O(M^{-1/2})$ finite-sample discrepancy rather than a modelling error: in the population the two sides are identically equal. It degrades to 83% half a unit away, which is the behaviour BHHH is known for and the reason the trust region is the safeguard.

6.2 The four questions

Table 2: Every arm, scored identically. f and $\|\nabla f\|$ are over the whole population at the stopping point. *terms* is the cumulative count of individual evaluations, which is the cost; iterations are not. **max_it.** and **first_ord.** abbreviate the statuses **max_iter** and **first_order**.

arm	iters	status	$N_{\min}$	$N_{\max}$	terms	f (true)	$\ \nabla f\ $	acc
raw, VAI	300	max_it.	10	3946	1.982×10^4	1.510823	3.284×10^{-1}	0.6509
Naive(0.75,2), VAI	300	max_it.	50	4717	9.794×10^5	0.782861	2.683×10^{-2}	0.6924
monotone ($a = 1$), VAI	120	first_ord.	100	100000	1.725×10^6	0.781998	9.002×10^{-8}	0.6915
Naive, VAI/VAC	26	first_ord.	50	100000	9.627×10^5	0.781998	7.107×10^{-8}	0.6915
Naive, VAC	28	first_ord.	50	100000	1.019×10^6	0.781998	2.926×10^{-8}	0.6915
Naive <i>tv</i> , VAI/VAC	26	first_ord.	50	100000	9.627×10^5	0.781998	7.107×10^{-8}	0.6915
Naive <i>sHs</i> , VAI/VAC	24	first_ord.	50	100000	8.304×10^5	0.781998	4.736×10^{-7}	0.6915
Naive, VAI/VAC, exact H	31	first_ord.	50	100000	1.064×10^6	0.781998	1.210×10^{-9}	0.6915

1. Is the raw certified size unstable? Yes, and more sharply than “unstable” conveys. The size sits at $N_{\min} = 10$ for 79% of the run, and the transition ratio N_{k+1}/N_k ranges over $[0.0025, 263.07]$, with 9 of 299 transitions jumping by more than fourfold. The cause is arithmetic and not numerical: (3) reads $\hat{\sigma}$ and $\hat{\delta}$ off the *same* step, and when the decrease nearly vanishes the requirement jumps by orders of magnitude with nothing to damp it.

The consequence is worse than an ugly plot. With ten individuals and ten parameters the batch problem is degenerate, so the run is not solving the problem: it ends at $f = 1.510823$ against $\ln 5 = 1.609438$ at the starting point, having moved almost nowhere in 300 iterations. It is also by far the cheapest arm, at 1.982×10^4 terms, which is what a run that does nothing costs.

2. Does the naive smoothing fix it? It improves the run by a long way and does not fix it. Naive(0.75,2) with the floor brings the true objective from 1.510823 to 0.782861 and the true gradient from 0.3284 to 0.02683, and the bound holds at all 299 transitions. But the arm still reports **max_iter**.

Figure 2. Three sample update functions

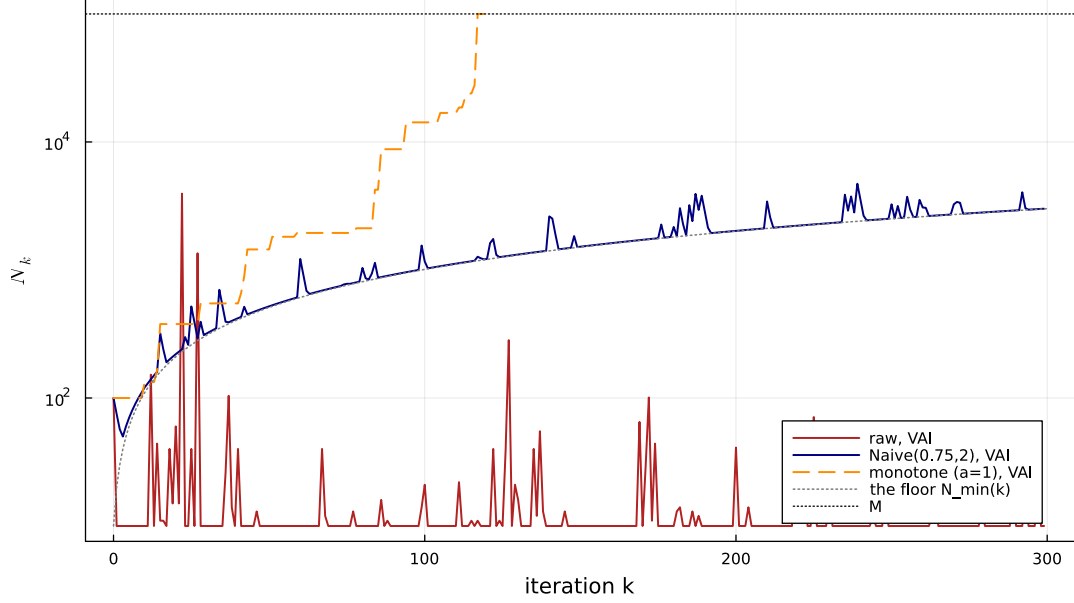


Figure 1: The three sample update functions under independent draws, on a logarithmic scale, with the rising floor $N_k^{\min} = 10 + 10k$ shown for reference. The raw rule sits on $N_{\min}$ and spikes; Naive(0.75, 2) tracks the floor; $a = 1$ climbs to the population and stays.

What converges under independent draws is the extreme setting $a = 1$: 120 iterations, and 1.725×10^6 terms, which is the most expensive arm in the table by 60%. So the lower bound is doing real work here, and $a = 0.75$ is not tight enough on its own on this problem. That is a qualification of the PreDoc’s report that Naive(0.75, 2.0) “empirically shows some better results”, not a contradiction of it: the PreDoc compares Naive against CMS-cum, and the monotone arm is neither.

3. Do the common-variable schemes converge in fewer iterations? Yes, and the gap is not marginal: 26 iterations under VAI/VAC and 28 under VAC, against an independent arm that does not converge in 300. This is the PreDoc’s central empirical claim about sample management and it reproduces.

Figure 3. VAI, VAI/VAC and VAC under the same rule

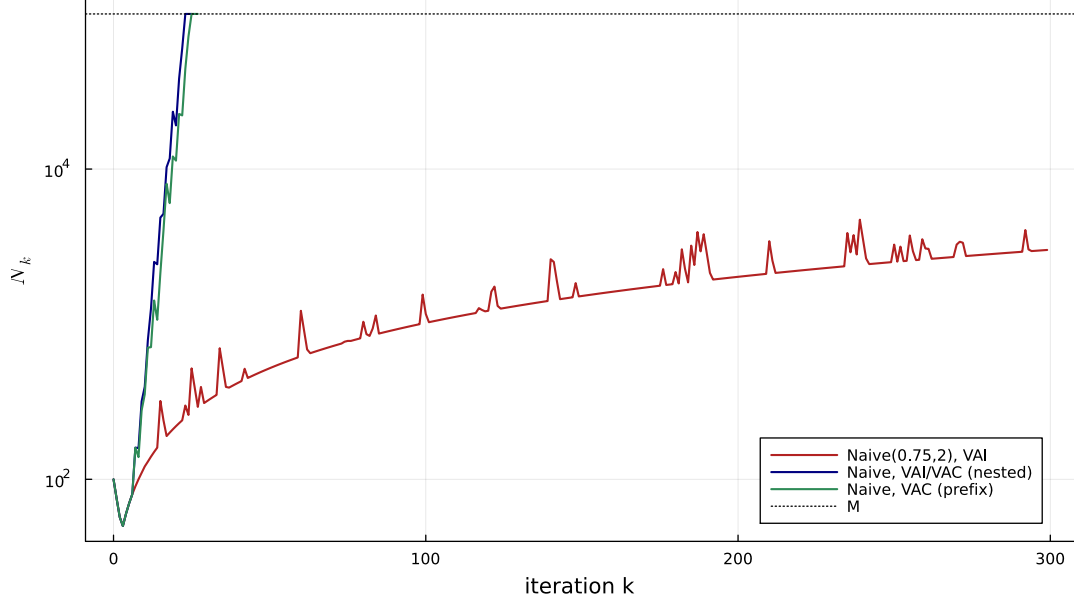


Figure 2: The same rule and the same seed under the three sample management schemes. The two common-variable arms reach the population and stop; the independent arm runs to the iteration cap without doing so.

Three qualifications follow from the same table.

First, the iteration count is not the cost. The three arms spend 9.794×10^5 , 9.627×10^5 and 1.019×10^6 terms, all within 6% of each other, because the converging runs reach the full population and their late iterations are correspondingly expensive. The defensible statement is not that common variables are cheaper. It is that on the same budget one arm converged and the other did not, which is a stronger claim and a different one.

Second, the mechanism is not confirmed. The PreDoc attributes the effect to common variables stopping $\tilde{f}$ from moving for reasons unrelated to the step. The natural diagnostic, the median of $|\tilde{f}_{k+1} - \tilde{f}_k|$ against the median achieved decrease over a common window, gives 1.430, 4.267 and 6.337 for VAI, VAI/VAC and VAC, which is the wrong order. The absolute movement does fall as predicted, from 7.605×10^{-2} to 6.796×10^{-3} and 8.638×10^{-3} , but so does the decrease, and by more. The comparison is confounded in any case: at a matched iteration index the VAI run is still far out taking large steps while the VAI/VAC run is nearly converged. We report the diagnostic and do not draw the conclusion from it.

Third, all three arms reach the same objective to six figures, 0.781998 against 0.782861, and the same accuracy to three. What separates them is whether they got there, not where they got.

The VAC hazard. The PreDoc warns that the prefix scheme can overfit its own prefix: while N_k is small the run minimises a different function, whose minimiser is some distance from the one sought, and the symptom is a spike then a fall in the sample size, possibly cycling.

A fall on its own is not the hazard, since with $a = 0.75$ the rule is permitted to fall and should whenever the decrease is large. The hazard is a fall taken while the *true* gradient is rising, and that is on the trace.

Table 3: Falls in the sample size, and whether the run was going backwards across them. The comparison that carries weight is the last two rows, which converged under the same rule and seed.

arm	iters	falls	falls with $\ \nabla f\ $ up	worst rise
Naive(0.75,2), VAI	300	56	33	$3.316\times$
Naive, VAI/VAC	26	6	1	$1.149\times$
Naive, VAC	28	8	4	$1.779\times$

The signature is present, and it is proportionally worse under VAC than under VAI/VAC, at 4 of 8 against 1 of 6 and a worst rise of $1.78\times$ against $1.15\times$. That is the ordering the PreDoc predicts, on runs identical in every other respect, and it is the mechanism it describes: VAC reuses one prefix while VAI/VAC refreshes it at the margin.

The VAI row has the largest raw count, and it is not evidence about sample management: a run that never converged has a gradient moving for its own reasons.

The hazard did not prevent convergence here, and it should not be expected to. The 100 000 individuals are generated independently, so the first hundred are an ordinary random sample. Sorted or clustered data is where it would bite, and this reproduction does not construct that case.

6.3 The two variance estimates

The earlier report could not make this comparison. Reconstructing $s^\top \tilde{B}s$ after the fact from an archived trace was refuted by its own check, on 84 of 300 iterations, for two independent reasons: the achieved reduction is not the difference of successive objective values on a rejected step, and under sampling the two values sit on different batches. Both estimates are now formed inside the iteration, on one batch at one iterate, and recorded.

Table 4: The two estimates against each other, over the 26 paired iterations of the VAI/VAC run.

iterations with both finite	26
median ratio $\hat{\sigma}_{sHs}^2/\hat{\sigma}_{tv}^2$	1.003150
ratio range	[0.999912, 1.218152]
iterations with ratio outside $[0.5, 2]$	0
negative outer-product values	0 of 26



Figure 3: The empirical and outer-product estimates of $\text{Var}(D_i)$, formed on the same batch at the same iterate of the VAI/VAC run, with $\|s_k\|$ on the same axes. The two are indistinguishable at this scale over most of the run, which is Table 4 seen as a trajectory.

The agreement is close: the median discrepancy is 0.3% and the worst is 22%. As a rule the choice makes little difference and slightly favours *sHs*, which converges in 24 iterations rather than 26 and at 8.304×10^5 terms rather than 9.627×10^5 .

Two cautions. The approximation is first order in $\|s_k\|$, so the spread in the ratio is the step length and not an estimation error; the notebook confirms the direction directly, with the ratio running $1.0059 \rightarrow 1.0008 \rightarrow 1.0001 \rightarrow 1.0000$ as $\|s\|$ falls through four decades. And the close agreement is a property of this problem, where Table 1 shows the identity holding to 1.2%. It should not be carried to a model where it does not.

6.4 BHHH against the exact Hessian

Table 5: The two model Hessians under the nested scheme, everything else fixed.

model	iters	terms	$\ \nabla f\ $ (true)	CG iterations
BHHHModel	26	9.627×10^5	7.107×10^{-8}	55
ExactHessian	31	1.064×10^6	1.210×10^{-9}	53

BHHH reaches the same objective in fewer iterations and at lower cost, and the exact Hessian reaches a smaller final gradient. Neither is a surprise at $p = 10$ with the identity holding to 1.2%: the two models are close, and the difference is within what one seed shows.

The cost argument is about p and not about this table. The exact Hessian costs $O(NJp^2)$ per batch against $O(NJp)$ for the score matrix, and the score matrix is computed for the gradient anyway, so the outer product is obtained free. At $p = 10$ the constant is small enough that the two run in comparable time.

One reporting note. `neval_hprod` reads 0 for the BHHH arm, because `OuterProductOperator` forms Bv from the score matrix without going through the model’s `hprod!`, so the counter never sees it. It did one product per CG iteration. The CG line is the one to read when the model is an outer product.

7 What is not reproduced

Four things, stated so that the reproduction is not read as more complete than it is.

1. **CMS and CMS-cum smoothing.** The PreDoc compares Naive against CMS-cum, and only Naive is implemented. CMS holds the size constant for $m = \lceil \epsilon_k^\beta / \tilde{\mu}_k \rceil$ iterations, which needs the rule to carry a countdown across iterations rather than answer from the current state. That is a different shape of rule, not a different constant, and `SmoothedSize` is not it.
2. **A separate Hessian sample $\mathcal{H}_k$.** The PreDoc’s Step 1 draws $\mathcal{H}_k$, possibly non-uniformly, and then assumes $\mathcal{H}_k \subset \mathcal{S}_k$ throughout. Here the model is built on the gradient batch, which is the case $\mathcal{H}_k = \mathcal{S}_k$.
3. **The generating noise.** Discussed above: the keyword exists and the runs use `noise = 0`. The two readings of the PreDoc’s sentence are not reconciled.
4. **A convergence criterion.** The PreDoc states none, and its iteration counts are read off figures. The cap of 300 and the tolerance of 10^{-6} are ours. This is why comparisons against arms reporting `max_iter` are phrased as failures to converge within the cap.

8 Where things are

notebook	notebooks/Sampling/predoc_logit_v2.ipynb
driver	_audit/nb7_predoc_logit_mnl.jl
MultinomialLogit	src/Likelihood/models.jl
SmoothedSize, paired_variance_kind	src/Sampling/rules.jl
schemes, paired_op_variance	src/Sampling/oracles.jl
recording both variances	src/Trust-region/common.jl
superseded report	report/task14-predoc-logit/

The notebook is six figures and eight tables, executed end to end with sequential execution counts and no errors. Nothing in this report is computed here; every number is transcribed from that execution.